

(Somaiya Vidyavihar University)

Batch: A2 Roll No.: 16010324044

Experiment No :- 5

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

TITLE: Sampling rate conversion of Discrete time signals.

AIM: 1) To understand the concept of increasing / decreasing the sampling rate of a discrete time signals using up sampling / down sampling.

2) To understand the difference between up sampling and down sampling.

OUTCOME: Analyse and design sampling rate converter, Understand concept of Multi-rate signal processing.

PROCEDURE:

1. Consider a discrete time signal of twenty samples.
2. Observe the spectrum of the signal.
3. Up sample the signal by a factor of 2 OR 3 and observe the effect on the signal and its spectrum.
4. Use a suitable anti-imaging filter to filter the signal and then observe the effect on the signal and the spectrum.
5. Down sample the signal by a factor of 2 OR 3 and observe the effect on the signal and its spectrum.
6. Use suitable anti-aliasing filter to filter the original signal before down sampling and then observe the effect on the signal and the spectrum.

OBSERVATIONS

1. Up sampling of discrete time signals

Original sampling rate = 1000 Hz

Length of the original signal = 20 samples

Up sampling factor = 3

Length of the up sampled signal = 60 samples

Effect on the spectrum after up sampling = Spectral images are introduced at higher frequencies.

2. Down sampling of discrete time signals

Original sampling rate = 1000 Hz

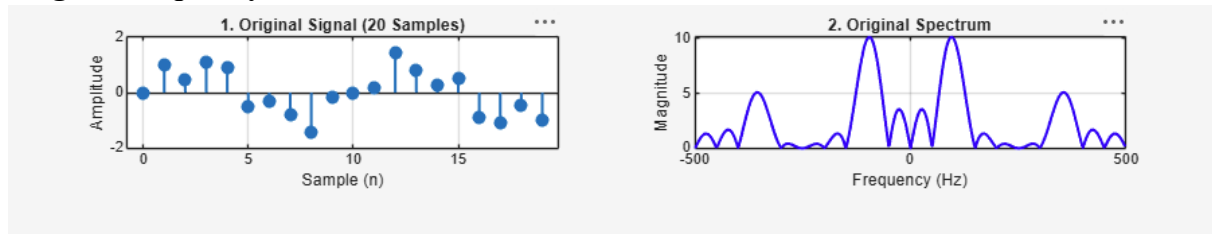
Length of the original signal = 20 samples

Down sampling factor = 2

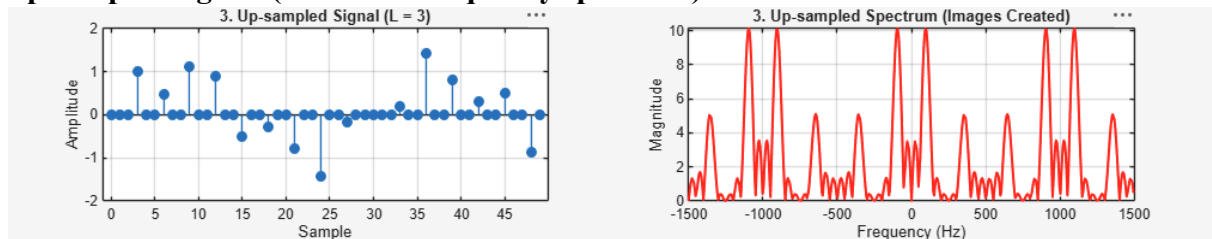
Length of the down sampled signal = 10 samples

Effect on the spectrum after down sampling = Aliasing occurs because the 350 Hz component exceeds the new Nyquist frequency of 250 Hz.

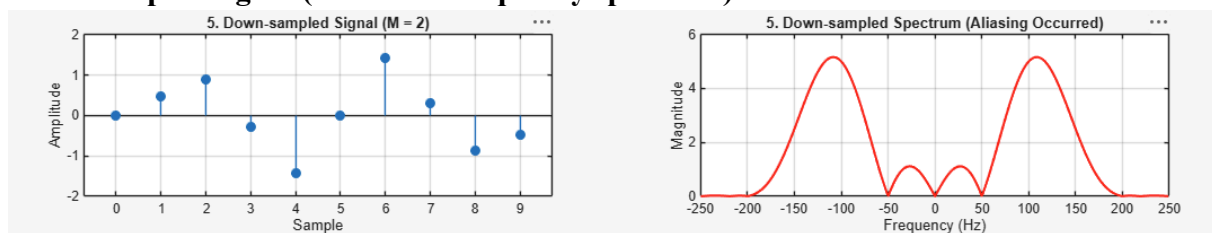
Original frequency



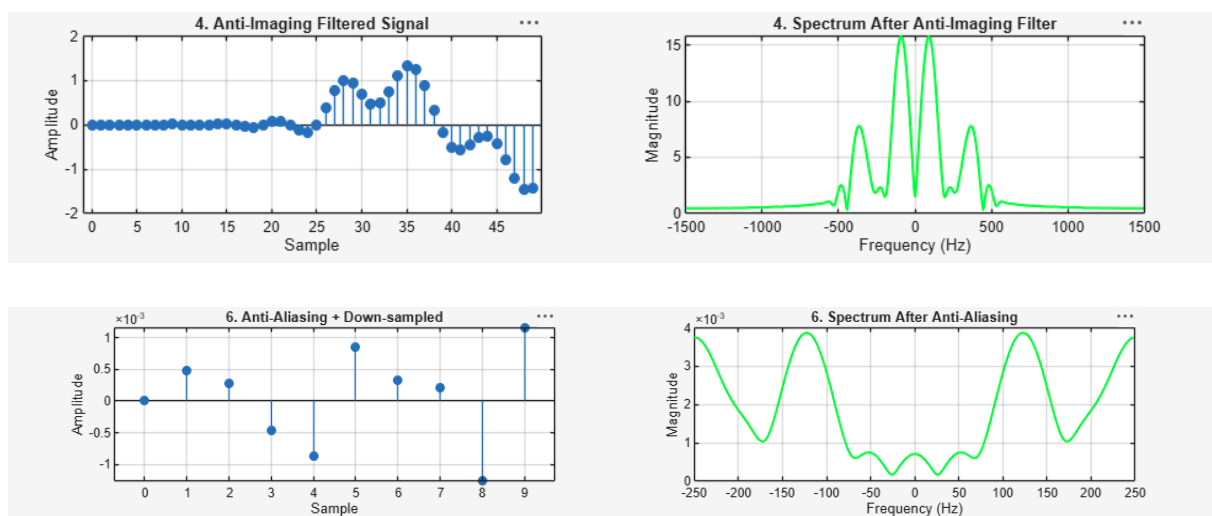
Up sampled signal (Time and frequency spectrum)



Down sampled signal (Time and frequency spectrum)



Interpolated signal:



CONCLUSION

In this experiment, I successfully performed up-sampling, interpolation, and down-sampling of discrete-time signals. Up-sampling increased the number of samples and introduced spectral images, which were removed using an anti-imaging filter. Down-sampling reduced the number of samples and caused aliasing when frequency components exceeded the new Nyquist frequency; this effect was reduced using an anti-aliasing filter. Thus, the effects of sampling-rate conversion and the importance of suitable filtering in multi-rate signal processing were studied successfully.

Signature of faculty in-charge